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RiskLens AI

AI-Powered Financial Risk Intelligence for Smarter, Safer Investments



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The Problem & Our Approach

RiskLens AI

The Problem

- Financial markets generate **massive amounts** of news, economic announcements and corporate disclosures.
- Manually interpreting this information and understanding its **potential impact** on a portfolio is challenging.
- A news headline alone is **not enough** — we need to know: What does it mean? Which assets are affected?
- How can we **quantify the risk** before it impacts our portfolio?

From News to Actionable Insights

Unstructured Financial News
News articles, headlines, economic updates

→

Structured Risk Signals
Sentiment, event category, impact score

→

Portfolio Impact Analysis
Understanding and managing potential risk

Our Solution

RiskLens AI connects **financial news intelligence** with **portfolio stress testing** to turn unstructured information into **structured risk insights.**

- Ingest & Analyze Financial News
- Assess Portfolio Exposure
- Run Scenario-Based Stress Tests
- Make Informed Risk Decisions

03 | System Design – End-to-End Architecture

RiskLens AI

Financial News Sources

- GDELT
- Financial RSS
- Synthetic Replay



Ingestion & Deduplication

- Fetch & Clean Data
- Remove Duplicates
- Store for Analysis

NLP Risk Engine

FinBERT (Sentiment Analysis)

+

Lexicon Fallback (Loughran–McDonald)

- Classify Events
- Estimate Impact Score

Structured Risk Signals

- Sentiment Score
- Event Category
- Impact Level
- Confidence

Portfolio & Scenario Mapping

- Map to Portfolio Exposure
- Choose Scenarios (e.g. Rate Hike, Credit Event)

Stress Testing & Monte Carlo Analysis

- Estimate Portfolio Loss
- Compute VaR / CVaR
- Analyze Risk Impact

RiskLens AI Dashboard

- Visualize Results
- Track Simulations
- Support Informed Decisions

Real-Time Insights for Smarter Financial Decisions

Technology Stack

Frontend
React 19 + TypeScript (Vite, Recharts)

Backend
Python + FastAPI (Pydantic)

Database
SQLite + SQLAlchemy

NLP Models
FinBERT + Loughran–McDonald

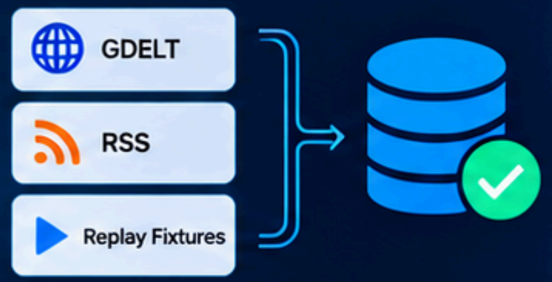
Risk Analytics
Monte Carlo Simulation, VaR/CVaR

04 Implementation Highlights

Turning Financial News into Actionable Risk Intelligence

RiskLens AI

1 Financial News Ingestion & Deduplication



- Fetch news from GDELT, RSS feeds and synthetic replay data
- Remove duplicates & clean data
- Store for downstream analysis

Why this? Ensures high-quality, **reliable** financial information.

2 NLP Sentiment Analysis & Event Classification



- Analyze sentiment using FinBERT
- Classify events (e.g. rate hike, credit event, market risk)
- Compute impact score & confidence

Why this? Combines deep **learning** with financial domain knowledge.

3 Portfolio & Scenario Mapping



- Bonds
- Equities
- Loans
- Options

- Link risk signals to portfolio exposures
- Map scenarios (rate hike, credit default, market shock)
- Evaluate asset-level impact

Why this? Connects news-driven risks to **real portfolio holdings**.

4 Stress Testing & Monte Carlo Risk Analysis



- VaR
- CVaR
- Loss Estimate

- Run economic scenarios on the portfolio
- Simulate outcomes using Monte Carlo methods
- Get quantified loss estimates & risk metrics

Why this? Helps users **prepare** for potential market shocks.

Tech Stack Powering It All

- React + TypeScript
- Python + FastAPI
- SQLite + SQLAlchemy
- FinBERT + Loughran–McDonald
- Recharts & Data Analytics

*A robust, modular and scalable implementation for **intelligent financial risk analysis**.*

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Key Results

From financial news to quantified portfolio risk

RiskLens AI

1 Financial Intelligence

- Financial news ingestion from supported sources and replay fixtures.
- FinBERT sentiment analysis with a lexicon-based fallback.
- Structured event categories, sentiment scores and impact estimates.

News & Analysis

Central bank signals higher rates ahead amid inflation concerns 2h ago • Reuters	Negative	-0.72
Tech stocks gain on stronger earnings outlook 4h ago • Bloomberg	Positive	0.68
Global markets react to geopolitical tensions 6h ago • CNBC	Neutral	0.02

Article Analyzer

Fed signals potential rate hike in next meeting

The Federal Reserve indicated it may consider another rate hike in its upcoming meeting, citing persistent inflationary pressures...

Sentiment (FinBERT)	Negative	-0.72
Fallback (Lexicon)	Negative	-0.65
Event Category	Monetary Policy	
Impact Estimate	High	



2 Portfolio Risk Analysis

- Multi-asset portfolio covering loans, bonds, equities and a put-option hedge.
- Scenario-based valuation and portfolio loss decomposition.
- Illustrative Monte Carlo VaR and CVaR estimates.

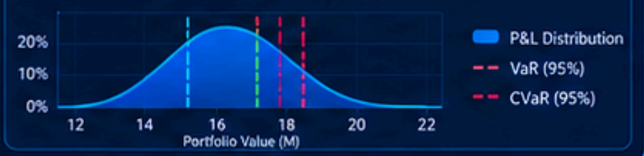
Portfolio Management



Stress Testing

VaR (95%)	CVaR (95%)	Daily P&L
$695,350	$1,034,520	+ $32,460

Portfolio Loss Distribution





3 Demonstrated End-to-End Workflow

- Integrated dashboard and persistent simulation records.
- Distinct provenance labels for model inference, fallback analysis and synthetic replay.



Portfolio Baseline
$16.99M

Illustrative Rate-Hike Loss
$695,350
4.09% of baseline

06 | Domain Impact | Turning financial information into **actionable risk insights**



Faster news triage

Helps analysts organize financial news into structured sentiment and event-risk signals.

Exposure-aware analysis

Connects potential financial events with relevant portfolio holdings and asset classes.

Scenario-based decision support

Estimates how hypothetical economic shocks could affect portfolio value and potential losses.

Traceability

Maintains analysis and simulation records to support review of the workflow and its assumptions.

Example Use Case

Central-bank rate hike

A rate increase may **reduce the market value** of existing fixed-rate bonds and **raise borrowing costs** for some corporate borrowers. RiskLens AI allows an analyst to explore the **estimated portfolio impact** under a configured scenario.

RiskLens AI connects what happened in the news with **which exposures** may be affected and what **a hypothetical shock could mean financially**.

07 | Limitations & Next Steps

Transparent assumptions, stronger validation, **greater realism**

Current Limitations

- Synthetic replay data is used for **reproducible demonstrations.**
- Monte Carlo correlation inputs are illustrative, **not fully calibrated to historical market data.**
- Options valuation uses a simplified **delta-only approach.**
- Risk estimates depend on input quality and **scenario assumptions.**
- The prototype is not a production trading **or regulatory-capital system.**

Next Steps

- Calibrate risk models using **historical market data.**
- Improve options valuation to capture **nonlinear risks.**
- Expand scenario coverage and **model validation.**
- Evaluate NLP outputs on labelled **financial-news datasets.**
- Strengthen monitoring, **auditability** and **data-source validation.**



Our Direction

Build on an integrated **financial-news-to-risk workflow** with better-calibrated models, richer valuation and stronger empirical validation.